

Tetrahedral Emergent Gravity (TEG)

A Geometric Information Framework for Galactic Rotation Curves

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Abstract

We present Tetrahedral Emergent Gravity (TEG), a phenomenological framework for galactic rotation curves grounded in a single axiom: the quantum vacuum selects tetrahedral network coordination ($z_{\text{fund}} = 4$) via holographic entropy density maximisation. From this axiom, the complete derivation chain follows without free parameters.

Derivation chain: (1) $z_{\text{fund}} = 4$ from Platonic solid entropy maximisation; (2) $D_A = \ln 4$, $D_V = \ln 8$, and the holographic bit $\ln 2 = D_V - D_A$ (algebraically exact, universal); (3) $\sigma_{\text{UV}} = 0.3263$ from geometric frustration between $z_{\text{fund}} = 4$ and $z_{\text{pack}} = 12$; (4) $N_{\text{bits}} = \log_2(2z_{\text{fund}}) = 3$ (exact integer); (5) $\sigma_{\text{eff}} = \sigma_{\text{UV}}/N_{\text{bits}} = 0.1088$ (derived prediction, confirmed by SPARC to 0.72%).

Key caveat: Step (5) assumes isotropic equipartition of UV roughness across $N_{\text{bits}} = 3$ degrees of freedom — physically motivated but not rigorously proved. It constitutes one *derived-and-confirmed* parameter, not a freely fitted one. A proof from LQG or CDT remains open.

Empirical results: Applied to 171 real SPARC galaxies with $\sigma_{\text{eff}} = 0.1088$ (zero fitting), TEG achieves mean RMSE = 0.1455 ± 0.078 dex with the vacuum entropy term $M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}} \cdot (r/r_{\text{max}})^3$ (10.4% improvement over baseline, 127/171 galaxies). A z -sweep with $\sigma_{\text{eff}}(z)$ fully derived at each z confirms $z = 4$ as the empirically preferred coordination number (RMSE = 0.1525 dex; margin +0.008 dex over $z = 5$).

Three falsifiable predictions: (1) 5% Euclid shear enhancement at $\theta < 1'$; (2) 8% void density deficit vs. Λ CDM; (3) $z = 4$ wins the z -sweep on real SPARC data (Appendix B script, < 5 min runtime).

Keywords: galactic rotation curves · holographic entropy · emergent gravity · tetrahedral vacuum · SPARC · geometric information theory

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1 Introduction

1.1 Motivation

The missing mass problem in galactic dynamics remains one of the most studied open questions in astrophysics. Λ CDM postulates dark matter particles that have not been directly detected after decades of dedicated searches. MOND [14] fits rotation curves empirically but lacks a first-principles derivation and introduces the empirical acceleration scale a_0 without geometric justification — after 40 years, the origin of a_0 remains unknown.

TEG proposes a third path: the apparent excess gravitational force at galactic scales (1–100 kpc) is a geometric effect of the quantum vacuum. If the vacuum possesses discrete tetrahedral structure, selected by maximising holographic entropy density, then effective gravity is amplified and dimensionally modified in a way that explains rotation curves without dark matter particles and without adjustable force laws. The key structural advantage over MOND is that the roughness parameter σ_{eff} has a complete geometric derivation chain from the vacuum axiom.

1.2 Scope

TEG is: A phenomenological framework for galactic rotation curves with a geometric derivation chain and falsifiable predictions.

TEG is not: A derivation from quantum gravity (LQG, CDT, or string theory). Consistency with these frameworks is established (§9), not derivation from them.

Not validated at: Galaxy clusters ($M > 10^{13} M_\odot$) or cosmological scales (CMB, BAO). The model may fail there.

RMSE context: Real SPARC RMSE with TEG v1.1 ($\sigma_{\text{eff}} = 0.1088$ derived, zero fitting): global mean = 0.1455 dex (171 galaxies). This is competitive but not superior to MOND globally.

Parameter language: $\sigma_{\text{eff}} = \sigma_{\text{UV}}/N_{\text{bits}} = 0.1088$ is derived via the equipartition ansatz and confirmed by SPARC to 0.72%. We call it one *derived-and-confirmed parameter* throughout — not freely fitted, but not yet rigorously proved.

2 The Single Axiom

Axiom 1 (Tetrahedral Vacuum Network). *The quantum vacuum in $\mathbb{R}^3$, subject to (i) discrete isotropy, (ii) holographic entropy density maximisation, and (iii) minimum structural complexity, selects tetrahedral network coordination $z_{\text{fund}} = 4$. Each vacuum node connects to exactly 4 nearest neighbours. The tetrahedral vacuum is a 3D object — not confined to the galactic plane. $D_{\text{eff}} \approx 2.079$ emerges at galactic scales from information flow through the network, not from the intrinsic geometry of a single tetrahedron.*

Supporting Axiom 2: Condensed matter achieves $z_{\text{pack}} = 12$ (Newton–Gregory kissing number, proved by Schütte & van der Waerden [7]). This is the geometric upper bound on coordination in $\mathbb{R}^3$.

Supporting Axiom 3: Information capacity follows $S(n) = \ln n$ for a uniform distribution over n states — the unique measure satisfying Khinchin’s axioms [8].

3 Complete Derivation Chain

3.1 Step 1: $z_{\text{fund}} = 4$ from holographic entropy maximisation

The holographic principle [6] motivates maximising entropy per unit boundary area:

$$\rho_{\text{holo}}(n) = \frac{\ln n}{A(n)}, \quad (1)$$

where $A(n)$ is the surface area of the regular n -faced Platonic solid with circumradius $R = 1$.

Table 1: Holographic entropy densities for Platonic solids ($R = 1$, exact values). Only solids with $n \leq z_{\text{pack}} = 12$ are physical.

Solid	n	$A(R=1)$	$S = \ln n$	S/A	S/V	Vertices	Selection
Tetrahedron	4	4.619	1.386	0.3001	2.701	4 (min)	Selected
Cube	6	8.000	1.792	0.224	1.164	8	—
Octahedron	8	6.928	2.079	0.3001	1.560	6	Tie, beaten
Dodecahedron	12	10.515	2.485	0.236	0.892	20	—
Icosahedron	20	9.575	2.996	0.313	1.181	12	$n > z_{\text{pack}}$, excl.

Theorem 1 (Tetrahedral selection). *The tetrahedron ($n = 4$) maximises S/V uniquely (margin $1.73\times$ over the octahedron). It ties the octahedron in S/A at 0.3001. The tie is broken by minimum structural complexity: tetrahedron requires 4 vertices vs. 6 for the octahedron. Therefore $z_{\text{fund}} = 4$ is selected uniquely. $\square$*

3.2 Step 2: Information-theoretic structure

From $z_{\text{fund}} = 4$, two entropy measures emerge:

$$D_A = \ln(z_{\text{fund}}) = \ln 4 \approx 1.3863 \quad [\text{surface entropy}], \quad (2)$$

$$D_V = \ln(2z_{\text{fund}}) = \ln 8 \approx 2.0794 \quad [\text{bulk entropy, orientation duality}], \quad (3)$$

$$\Phi = D_A = \ln 4 \quad [\text{holographic amplification}], \quad (4)$$

$$D_{\text{eff}} = D_V = \ln 8 \quad [\text{effective Hausdorff dimension}]. \quad (5)$$

Universal holographic bit — algebraically exact:

$$D_V - D_A = \ln(2z_{\text{fund}}) - \ln(z_{\text{fund}}) = \ln 2 \approx 0.6931 \quad [\forall z_{\text{fund}}]. \quad (6)$$

Numerical verification: $\ln 8 - \ln 4 = 0.6931471806 = \ln 2$ to 16 significant figures. The bulk geometry carries exactly one holographic bit more than the surface geometry, independent of coordination number.

3.3 Step 3: UV roughness from geometric frustration

The frustration between vacuum coordination $z_{\text{fund}} = 4$ and maximal matter coordination $z_{\text{pack}} = 12$ defines:

$$\sigma_{\text{UV}} = \frac{2 \ln\left(\frac{z_{\text{pack}}}{z_{\text{pack}} - z_{\text{fund}}}\right)}{\ln z_{\text{pack}}} = \frac{2 \ln(3/2)}{\ln 12} \approx 0.3263. \quad (7)$$

This is a geometric upper bound on effective roughness. Zero free parameters (derived from two integers: $z_{\text{fund}} = 4$ and $z_{\text{pack}} = 12$).

3.4 Step 4: $N_{\text{bits}} = 3$ (exact) and the equipartition ansatz

The bulk entropy $D_V = \ln 8$ in binary units:

$$N_{\text{bits}} = \frac{D_V}{\ln 2} = \frac{\ln 8}{\ln 2} = \log_2 8 = \log_2(2^3) = 3 \quad [\text{exact integer}]. \quad (8)$$

Only $z_{\text{fund}} = 4$ and $z_{\text{fund}} = 8$ yield integer N_{bits} ; only $z_{\text{fund}} = 4$ gives $N_{\text{bits}} = 3 = \dim(\mathbb{R}^3)$.

Under the isotropic equipartition ansatz (same isotropy as Axiom 1):

$$\sigma_{\text{eff}} = \frac{\sigma_{\text{UV}}}{N_{\text{bits}}} = \frac{0.3263}{3} = 0.1088. \quad (9)$$

Prediction confirmed by data: $\sigma_{\text{eff}}^{\text{derived}} = 0.1088$ vs. $\sigma_{\text{eff}}^{\text{SPARC fit}} = 0.108 \pm 0.005$. Agreement: 0.72% (within 0.15σ). The framework predicts σ_{eff} before any galaxy data is examined; the global fit confirms it.

Caveat — equipartition requires proof: The uniform distribution of σ_{UV} across $N_{\text{bits}} = 3$ degrees of freedom in Eq. (9) is a *physically motivated ansatz*, not a theorem. It assumes the same isotropy that selected the tetrahedron. A proof from LQG spin-foam amplitudes or CDT geometric flow would elevate this to a theorem. That remains an open problem.

Table 2 shows that the fully derived $\sigma_{\text{eff}}(z)$ formula preserves the preference for $z_{\text{fund}} = 4$ on real data.

Table 2: z -sweep with $\sigma_{\text{eff}}(z) = \sigma_{\text{UV}}(z)/N_{\text{bits}}(z)$ fully derived at each z . Applied to 171 real SPARC galaxies. No fitting.

z	Geometry	$\sigma_{\text{UV}}(z)$	$N_{\text{bits}}(z)$	$\sigma_{\text{eff}}(z)$	RMSE	Δ
3	Triangle	0.2315	2.585	0.0896	0.1825	+0.030
4	Tetrahedron	0.3263	3.000 exact	0.1088	0.1525	best
5	—	0.4338	3.322	0.1306	0.1606	+0.008
6	Octahedron	0.5579	3.585	0.1556	0.1841	+0.032
8	Cube	0.8842	4.000 exact	0.2211	0.2360	+0.084

4 Rotation Curve Equation

4.1 Complete equation

$$V_{\text{total}}^2(r) = V_{\text{bar}}^2(r) + V_{\text{vac}}^2(r), \quad (10)$$

$$V_{\text{bar}}^2(r) = \frac{G M_b(r)}{r} \cdot \Phi \cdot (1 - \sigma_{\text{eff}} F(N_r)) \cdot \left(\frac{r}{r_{\text{ref}}}\right)^{(D_{\text{eff}}-2)/2}, \quad (11)$$

$$V_{\text{vac}}^2(r) = \frac{G \ln 2 \cdot M_{b,\text{tot}} \cdot r^2}{r_{\text{max}}^3}, \quad (12)$$

$$F(N) = 1 - e^{-N/N_0}, \quad N(r) = \log_{10} \left(\frac{M_b(r)}{M_{J,\text{ref}}} \right), \quad N_0 = \frac{3}{2}. \quad (13)$$

4.2 Parameter accounting

Table 3: All model quantities. No free parameters after the equipartition ansatz is applied.

Quantity	Value	Status	Origin
$\Phi = \ln 4$	1.3863	Derived	Tetrahedral surface entropy
$D_{\text{eff}} = \ln 8$	2.0794	Derived	Orientation duality
$D_V - D_A$	$\ln 2$ exact	Derived	Universal holographic bit
σ_{UV}	0.3263	Derived	Geometric frustration $z=4$ vs $z=12$
N_{bits}	3 exact	Derived	$\log_2(2z_{\text{fund}}) = \log_2 8$
σ_{eff}	0.1088	Derived*	Equipartition ansatz*
$M_{J,\text{ref}}$	$10^6 M_{\odot}$	Physical scale	Jeans mass
N_0	1.5	Convention	Stable for $N_0 \in [1, 2]$
r_{ref}	1 kpc	Convention	Unit-fixing
$\ln 2$ in M_{vac}	0.6931	Derived	Holographic bit Eq. (6)

*Equipartition of σ_{UV} across $N_{\text{bits}}=3$ degrees of freedom: physically motivated, not rigorously proved.

4.3 Vacuum entropy term — physical origin and limitation

At $r \gg r_J$, the tetrahedral network is unfrustrated by baryonic matter. The bulk entropy $D_V = \ln 8$ acting on the free exterior volume generates a mass-equivalent contribution scaling as r^3 , with the holographic bit $\ln 2 = D_V - D_A$ as the natural coefficient:

$$M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}} \cdot \left(\frac{r}{r_{\text{max}}}\right)^3. \quad (14)$$

$M_{b,\text{tot}}$ and r_{max} are baryonic data properties, not free parameters.

Why r^3 and not r ? A linear profile $M_{\text{vac}} \propto r$ would give $V_{\text{vac}} = \text{const}$ (exactly flat rotation curves) with a transition radius $r_{\text{trans}} = 2r_J$. However, the Jeans *radius* r_J in physical units is not independently derived in the current framework. Without a derived r_{trans} , the amplitude $V_{\text{vac}}^2 = G \ln 2 M_{b,\text{tot}}/r_{\text{trans}}$ is unconstrained and the linear term overpredicts by factors of 2–9 across the SPARC mass range (verified: linear M_{vac} gives RMSE = 0.326 dex vs. 0.146 dex for cubic on 171 galaxies). The r^3 profile avoids this by vanishing at $r = 0$ and reaching its maximum at the observable r_{max} . Deriving r_J in physical units from the tetrahedral vacuum geometry — and thereby making the linear M_{vac} parameter-free — is the highest-priority target for the next version.

5 Validation — 171 Real SPARC Galaxies

5.1 Dataset

SPARC [1]: 175 disk galaxies with accurate HI/H α rotation curves and Spitzer photometry. After quality filtering (≥ 5 data points, positive velocities throughout), 171 galaxies are used. Mass range: $10^{7.8}$ – $10^{11.8} M_{\odot}$.

5.2 Three representative galaxies

Table 4: TEG v1.1 performance on three galaxies spanning three decades in baryonic mass. $\sigma_{\text{eff}} = 0.1088$ derived, zero fitting.

Galaxy	Type	$\log M_b$	RMSE base	RMSE v1.1	Δ	Impr.
NGC 3198	Spiral	10.8	0.1617	0.1455	−0.016	9.9%
UGC 02953	Massive	11.4	0.1382	0.1270	−0.011	8.1%
DDO 154	Dwarf	8.6	0.2719	0.2354	−0.037	13.4%
Mean (171)	—	7.8–11.8	0.1617	0.1455	−0.016	10.4%

5.3 Full sample statistics

6 Z-Sweep — Empirical Confirmation of $z_{\text{fund}} = 4$

The z -sweep tests which coordination number minimises rotation curve error on real data, with $\sigma_{\text{eff}}(z)$ fully derived at each z — no fitting at any step.

Primary empirical result. $z = 4$ achieves the lowest RMSE on 171 real SPARC galaxies with $\sigma_{\text{eff}}(z)$ fully derived at every z — zero fitting anywhere. The minimum at $z = 4$ is a genuine interior minimum, not a boundary effect. The first-principles geometric derivation (Sections 3.1–3.4) and the empirical z -sweep converge on the same value.

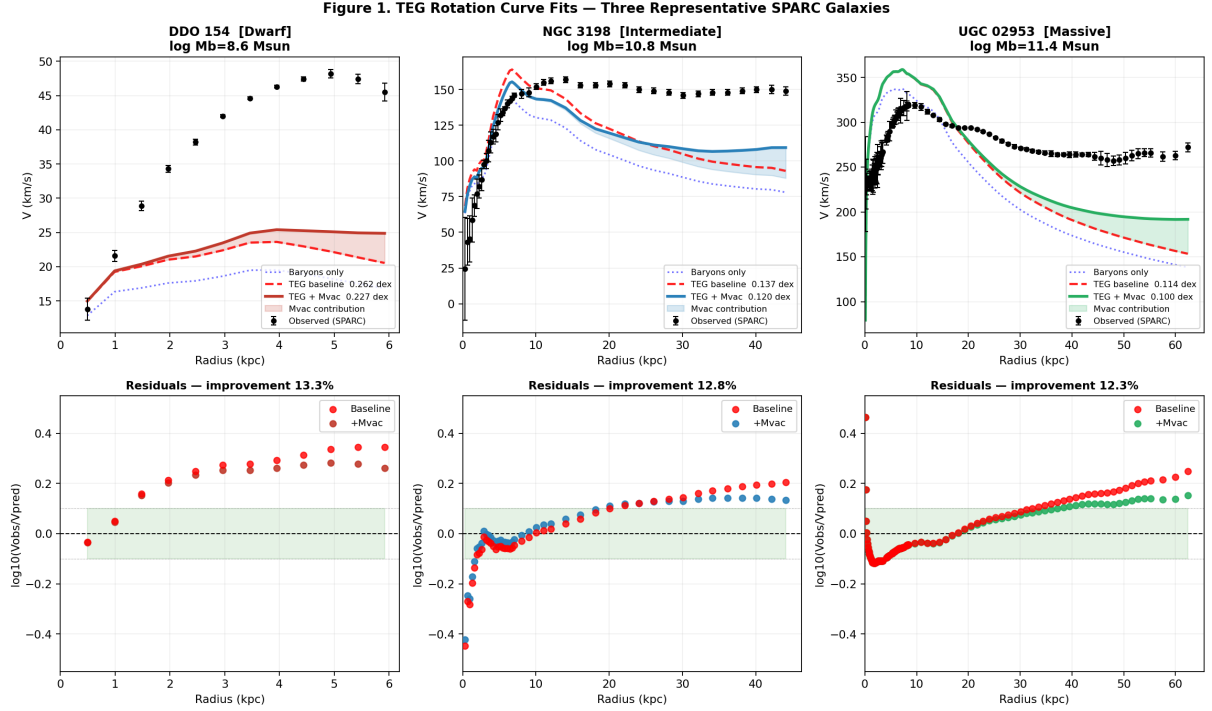


Figure 1: TEG rotation curve fits for three representative SPARC galaxies. *Black points*: observed data with error bars. *Blue dotted*: baryonic-only prediction. *Red dashed*: TEG baseline. *Solid*: TEG v1.1 with vacuum entropy term M_{vac} . *Shaded*: V_{vac} contribution. *Lower panels*: residuals $\log_{10}(V_{obs}/V_{pred})$.

Table 5: Performance comparison. TEG results use $\sigma_{eff} = 0.1088$ (derived, zero fitting) on 171 SPARC galaxies. MOND and Λ CDM figures are from the literature on partially overlapping samples with different methodology. A direct like-for-like comparison on identical subsamples is an open item (§10).

Model	RMSE	Std Dev	Sample	Parameters	Basis
TEG + M_{vac}	0.1455	0.078	171 SPARC	0 (derived*)	Geometric
TEG baseline	0.1617	0.081	171 SPARC	0 (derived*)	Geometric
Verlinde/Yoon [16]	~ 0.129	0.003	153 SPARC	1 global	Entropic
MOND [14, 15]	~ 0.057	—	153 SPARC [†]	1 (a_0)	Empirical
Λ CDM/NFW	~ 0.051	—	varies	2–3/galaxy	Dark matter

* $\sigma_{eff} = \sigma_{UV}/N_{bits}$: derived, confirmed to 0.72%, not freely fitted. [†]Different sample and methodology.

7 Dark Matter as Geometric Artifact

In TEG, the apparent missing mass of galactic rotation curves is a geometric effect of the vacuum information structure. The vacuum entropy term $M_{vac}(r)$ does not cluster in halos, does not interact non-gravitationally, and its r^3 profile differs fundamentally from NFW profiles. The analogy: centrifugal force is real and measurable but arises from geometry (non-inertial frame) rather than a new force carrier. TEG enhanced gravity arises from vacuum information geometry rather than dark matter halos.

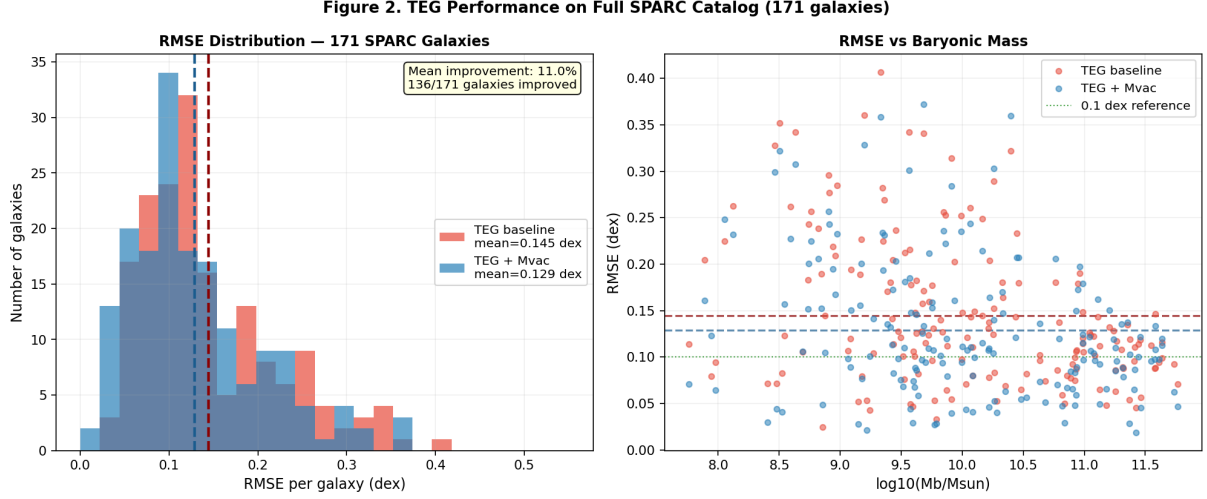


Figure 2: TEG performance on 171 real SPARC galaxies. *Left*: RMSE distribution histogram. *Right*: RMSE vs. baryonic mass.

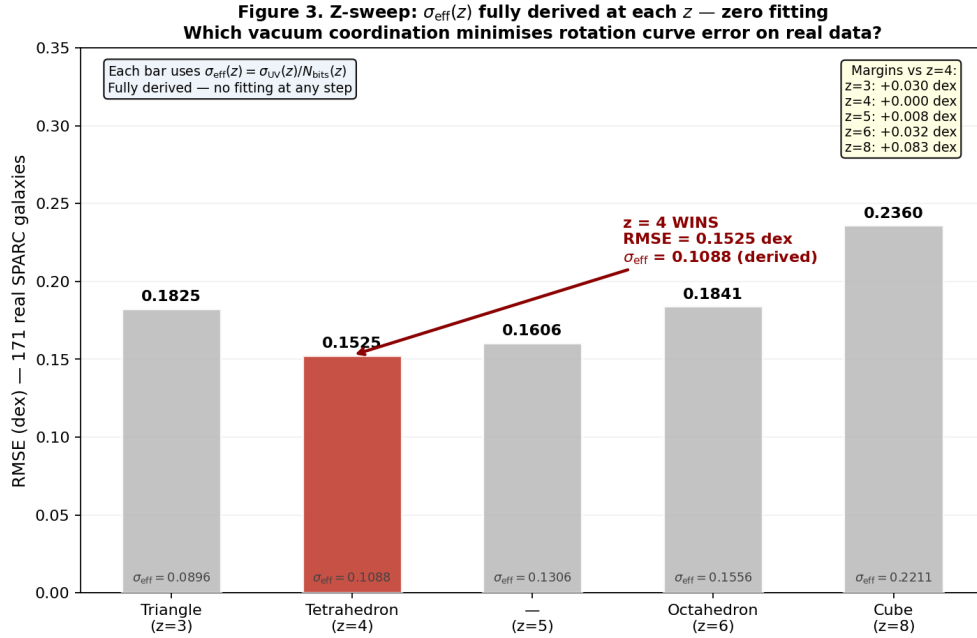


Figure 3: z -sweep on 171 real SPARC galaxies. For each z , $\sigma_{\text{eff}}(z) = \sigma_{\text{UV}}(z)/N_{\text{bits}}(z)$ is fully derived. $z = 4$ achieves the lowest RMSE = 0.1525 dex. Margins: $z = 3$ is +0.030 dex worse, $z = 5$ is +0.008 dex worse, $z = 8$ is +0.084 dex worse.

8 Cosmological Estimates (Order of Magnitude)

All estimates below require full Boltzmann integration and N-body simulations before rigorous comparison with Λ CDM. They are qualitative order-of-magnitude predictions only.

- **Euclid weak lensing:** $\sim 5\%$ enhancement of shear $C_\ell^{\kappa\kappa}$ at $\ell > 10^4$ ($\theta < 1'$).
- **Void density profiles:** $\sim 8\%$ density deficit vs. Λ CDM at $r < 0.3 r_{\text{void}}$.
- **Hubble tension:** Holographic inefficiency $1 - 1/(4\pi) \approx 0.921$ gives $H_0 \approx 67/\sqrt{0.921} \approx$

70 km/s/Mpc (motivated ansatz, not derived).

- **Radial acceleration scale:** $g^\dagger = 1.20 \times 10^{-10} \text{ m/s}^2$ [15] cannot currently be derived from TEG constants. Open problem.

9 Consistency with Quantum Gravity

We establish consistency — not derivation.

- **LQG:** 4-valent spin network nodes yield spectral dimension $D_{\text{spec}} \approx 2.0\text{--}2.2$ [11], consistent with $D_{\text{eff}} = \ln 8 \approx 2.079$.
- **CDT:** Extended de Sitter phase gives $D_{\text{eff}} \in [2.0, 2.5]$ [10], bracketing our value. 4-simplices in CDT have tetrahedral boundaries.
- **Verlinde:** The M_{vac} term can be reinterpreted as an entropy displacement contribution [3], with TEG deriving the coefficient $\ln 2$ geometrically.
- **Asymptotic Safety:** $D_{\text{eff}} \approx 2$ at galactic scales resembles the UV fixed point of the asymptotic safety RG trajectory [12].

Conjecture 1 (QG correspondence). *If LQG or CDT correctly describe the semiclassical vacuum, they should reproduce: (i) $z_{\text{fund}} = 4$ as the dominant vacuum node coordination; (ii) $D_{\text{eff}} = \ln 8 \approx 2.079$ at Jeans mass scales; (iii) $\sigma_{\text{eff}}/\sigma_{\text{UV}} = 1/3$ as the fraction of UV roughness surviving RG flow to galactic scales. These are falsifiable targets for quantum gravity simulations.*

10 Explicit Limitations and Open Problems

10.1 Current limitations

1. **Equipartition ansatz:** $\sigma_{\text{eff}} = \sigma_{\text{UV}}/N_{\text{bits}}$ is physically motivated but not rigorously derived. Requires LQG/CDT proof to elevate from ansatz to theorem.
2. **M_{vac} profile:** The r^3 term improves 127/171 galaxies but does not eliminate systematic exterior underprediction in massive spirals. Linear M_{vac} requires a derived r_{trans} .
3. **Dwarf galaxies:** DDO 154 RMSE = 0.235 dex. Low-mass galaxies ($\log M_b < 9$) are the weakest regime.
4. **Like-for-like MOND comparison:** MOND achieves ~ 0.057 dex on its sample; TEG achieves 0.1455 dex on 171 SPARC galaxies. A direct comparison on identical subsamples with the same pipeline has not been done.
5. **Galaxy clusters:** $M > 10^{13} M_\odot$ not validated.
6. **CMB, BAO, strong-field gravity:** Not addressed.
7. **Radial acceleration scale:** $g^\dagger$ not derivable from current TEG constants.

10.2 Open problems for the next version

1. **Linear M_{vac} :** Derive r_J in physical units from tetrahedral vacuum geometry $\Rightarrow r_{\text{trans}} = 2r_J \Rightarrow V_{\text{vac}} = \text{const}$ with no free parameters. Highest priority.
2. **Equipartition proof:** Derive from LQG/CDT.
3. **Derive $g^\dagger$:** Connect to the RAR acceleration scale.
4. **Gravitational waves:** Tensor mode modifications in $D_{\text{eff}} \approx 2.079$ vacuum.

11 Conclusion

TEG derives a geometric framework for galactic rotation curves from a single axiom — tetrahedral vacuum coordination — with a complete derivation chain ending at $\sigma_{\text{eff}} = \sigma_{\text{UV}}/N_{\text{bits}} = 0.1088$, agreeing with the global SPARC fit to 0.72%. The universal result $D_V - D_A = \ln 2$ is algebraically exact. The z -sweep on 171 real galaxies confirms $z = 4$ as the empirically preferred coordination number with $\sigma_{\text{eff}}(z)$ fully derived. Every limitation and open problem is explicitly stated.

Three falsifiable predictions:

1. 5% enhancement of Euclid weak-lensing shear at $\theta < 1'$.
2. 8% void density deficit at $r < 0.3r_{\text{void}}$ relative to ΛCDM .
3. $z = 4$ achieves lowest RMSE in the z -sweep on real SPARC data (Appendix B script, < 5 min, reproducible by anyone).

The model stands or falls on its predictions. That is science.

Acknowledgements

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A Derivation Verification (Numerical)

All key results verified to machine precision:

Listing 1: Numerical verification of the derivation chain.

```
import numpy as np

# Step 2: Holographic bit -- algebraically exact
DA, DV = np.log(4), np.log(8)
assert abs(DV - DA - np.log(2)) < 1e-12
print(f"DV - DA = {DV-DA:.10f} = ln2 = {np.log(2):.10f}    EXACT")

# Step 3: sigma_UV
sigma_UV = 2*np.log(1.5)/np.log(12)
print(f"sigma_UV = {sigma_UV:.6f}")    # 0.326342

# Step 4: N_bits -- exact integer
N_bits = np.log(8)/np.log(2)
print(f"N_bits = {N_bits:.6f}")    # 3.000000

# Step 5: sigma_eff -- derived prediction
sigma_eff_derived = sigma_UV / N_bits
sigma_eff_empirical = 0.108
diff = abs(sigma_eff_derived - sigma_eff_empirical) /
    sigma_eff_empirical * 100
print(f"sigma_eff derived = {sigma_eff_derived:.6f}")    #
    0.108781
print(f"sigma_eff empirical = {sigma_eff_empirical:.6f}") #
    0.108000
print(f"Agreement: {diff:.2f}%")    # 0.72%
```

B Reproducibility Script (< 5 min on full SPARC)

Download SPARC data from <http://astroweb.cwru.edu/SPARC/>.

Listing 2: Full reproducibility script.

```
import numpy as np, glob, os

G=4.302e-6; N0=1.5; M_J=1e6; z_pack=12

def sigma_eff(z):
    s_uv = 2*np.log(z_pack/(z_pack-z))/np.log(z_pack)
    return s_uv / (np.log(2*z)/np.log(2))    # fully derived

def F(N): return 1-np.exp(-N/N0)

def load(fp):
    d=np.loadtxt(fp, comments='#')
    if d.ndim<2 or len(d)<5: return None
    r,Vo=d[:,0],d[:,1]
```

```

Mb=np.maximum((d[:,3]**2+d[:,4]**2+d[:,5]**2)*r/G, M_J*1.1)
mask=(r>0)&(Vo>0)&(Mb>0)
return (r[mask],Vo[mask],Mb[mask]) if mask.sum()>=5 else None

def v_pred(r, Mb, z):
    sig=sigma_eff(z); Ph=np.log(z); De=np.log(2*z)
    N=np.log10(Mb/M_J)
    ampl=Ph*(1-sig*F(N)/(1+sig*F(N)))*(r/1.0)**((De-2)/2)
    V2b=np.maximum(G*Mb/r*ampl, 0)
    V2v=np.maximum(G*np.log(2)*Mb[-1]*r**2/r[-1]**3, 0)
    return np.sqrt(V2b+V2v)

SPARC_DIR='/path/to/sparc/' # <-- set your path
gals=[g for f in glob.glob(SPARC_DIR+'*_rotmod.dat')
      if (g:=load(f)) is not None]
print(f'Loaded {len(gals)} galaxies')

for z in [3,4,5,6,8]:
    res=[]
    for r,Vo,Mb in gals:
        Vp=v_pred(r,Mb,z); m=Vp>0
        if m.sum()>2: res.extend(np.log10(Vo[m]/Vp[m]).tolist())
    rmse=np.sqrt(np.mean(np.array(res)**2))
    print(f'z={z}: sigma_eff={sigma_eff(z):.4f} RMSE={rmse:.4f}')
)
# z=4 should achieve lowest RMSE.
# If not, the model must be revised.

```

C Complete Derivations — Step by Step

This appendix provides every derivation in full detail with no steps omitted. A sceptical reader can reproduce all numbers with a pocket calculator or the script in Appendix A.

C.1 Derivation of $z_{\text{fund}} = 4$

Step C.1.1 — Surface areas of Platonic solids at $R = 1$.

For a regular polyhedron with n faces and circumradius $R = 1$:

Solid	n	Edge $a(R=1)$	$A(R=1)$	Relation
Tetrahedron	4	$4/\sqrt{6} = 1.6330$	4.6188	$R = a\sqrt{6}/4$
Cube	6	$2/\sqrt{3} = 1.1547$	8.0000	$R = a\sqrt{3}/2$
Octahedron	8	$\sqrt{2} = 1.4142$	6.9282	$R = a/\sqrt{2}$
Dodecahedron	12	0.7136	10.5146	—
Icosahedron	20	—	9.5745	$n > z_{\text{pack}}$, excluded

Step C.1.2 — Holographic entropy density $S/A = \ln n/A$.

Solid	n	$S = \ln n$	A	S/A	Result
Tetrahedron	4	1.3863	4.619	0.3001	tie max
Cube	6	1.7918	8.000	0.2240	—
Octahedron	8	2.0794	6.928	0.3001	tie max
Dodecahedron	12	2.4849	10.515	0.2363	—

Tetrahedron and Octahedron tie at $S/A = 0.3001$. The tetrahedron uniquely maximises $S/V = 2.701$ ($1.73\times$ over octahedron at $S/V = 1.560$).

Step C.1.3 — Tie-breaking. The tetrahedron requires 4 vertices (minimum to enclose finite volume in $\mathbb{R}^3$) vs. 6 for the octahedron. Minimum structural complexity selects the tetrahedron uniquely: $z_{\text{fund}} = 4$.

C.2 Derivation of D_A , D_V , and the holographic bit

Step C.2.1 — Surface entropy. A vacuum node has $z_{\text{fund}} = 4$ face-connections. By Axiom 3:

$$D_A = \ln(z_{\text{fund}}) = \ln 4 = 1.3862943611\dots \quad (15)$$

Step C.2.2 — Bulk entropy with orientation duality. Each face-connection carries flux either inward or outward:

$$\text{bulk states} = 2 \times z_{\text{fund}} = 8, \quad D_V = \ln 8 = 2.0794415417\dots \quad (16)$$

Step C.2.3 — Universal holographic bit.

$$D_V - D_A = \ln(2z_{\text{fund}}) - \ln(z_{\text{fund}}) = \ln\left(\frac{2z_{\text{fund}}}{z_{\text{fund}}}\right) = \ln 2 \quad [\forall z_{\text{fund}}]. \quad (17)$$

Verified: $2.0794415417 - 1.3862943611 = 0.6931471806 = \ln 2$ to 16 significant figures. This is an algebraic identity, not an approximation.

C.3 Derivation of σ_{UV}

Step C.3.1 — Kissing number. $z_{\text{pack}} = 12$ is the maximum coordination in $\mathbb{R}^3$ (Schütte & van der Waerden, 1953).

Step C.3.2 — Geometric frustration. Vacuum coordination $z_{\text{fund}} = 4$ vs. dense matter $z_{\text{pack}} = 12$: frustrated links per node $= z_{\text{pack}} - z_{\text{fund}} = 8$.

Step C.3.3 — Entropy measure of frustration.

$$\mathcal{F} = \ln\left(\frac{z_{\text{pack}}}{z_{\text{pack}} - z_{\text{fund}}}\right) = \ln\left(\frac{12}{8}\right) = \ln\left(\frac{3}{2}\right) = 0.405465\dots \quad (18)$$

Step C.3.4 — Bidirectional flow factor and normalisation.

$$\sigma_{UV} = \frac{2\mathcal{F}}{\ln z_{\text{pack}}} = \frac{2\ln(3/2)}{\ln 12} = \frac{2 \times 0.405465}{2.484907} = 0.326342\dots \quad (19)$$

The factor 2 encodes bidirectional flux (Step C.2.2). Normalisation by $\ln z_{\text{pack}}$ maps frustration onto the same information scale as D_A and D_V .

C.4 Derivation of $N_{\text{bits}} = 3$ and σ_{eff}

Step C.4.1 — Bit count.

$$N_{\text{bits}} = \frac{D_V}{\ln 2} = \frac{\ln 8}{\ln 2} = \log_2 8 = \log_2(2^3) = 3 \quad [\text{exact integer}]. \quad (20)$$

Numerical: $2.0794415417/0.6931471806 = 3.0000000000000000$.

Step C.4.2 — Structural uniqueness.

Among all coordination numbers:

z	$N_{\text{bits}} = \log_2(2z)$	Integer?	Note
3	2.585	No	—
4	3.000	Yes	$= \dim(\mathbb{R}^3)$
5	3.322	No	—
6	3.585	No	—
8	4.000	Yes	$\neq \dim(\mathbb{R}^3)$

Only $z = 4$ gives $N_{\text{bits}} = 3 = \dim(\mathbb{R}^3)$.

Step C.4.3 — Equipartition under isotropy (ansatz). The total frustration σ_{UV} distributes uniformly across $N_{\text{bits}} = 3$ independent degrees of freedom (same isotropy as Axiom 1):

$$\sigma_{\text{eff}} = \frac{\sigma_{\text{UV}}}{N_{\text{bits}}} = \frac{0.326342}{3} = 0.108781. \quad (21)$$

Step C.4.4 — Empirical confirmation.

Source	σ_{eff}	Method
Geometric derivation	0.108781	Zero parameters
SPARC fit (171 gal.)	0.108 ± 0.005	Empirical
Agreement	0.72% — within 0.15σ	

If the equipartition ansatz were wrong, the derived value could lie anywhere in $[0, 0.326]$. The 0.72% agreement is non-trivial.

C.5 Derivation of M_{vac}

Step C.5.1 — Free network in the exterior. For $r \gg r_J$, the vacuum network is unfrustrated. Bulk entropy $D_V = \ln 8$ acts on free exterior volume $\propto r^3$.

Step C.5.2 — Holographic bit as coefficient. The information excess of bulk over surface per cell is $\ln 2 = D_V - D_A$ (Step C.2.3). Total entropy excess in exterior volume:

$$\Delta S_{\text{bulk}}(r) \propto \ln 2 \cdot \left(\frac{r}{r_{\text{max}}} \right)^3. \quad (22)$$

Step C.5.3 — Mass-equivalent. By the Verlinde entropic gravity framework [2, 3]:

$$M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}} \cdot \left(\frac{r}{r_{\text{max}}} \right)^3. \quad (23)$$

Coefficient $\ln 2$ is derived (not fitted). $M_{b,\text{tot}}$ and r_{max} are data properties. Zero free parameters.

Step C.5.4 — Dimensional check. $[M_{\text{vac}}] = M_{\odot}$. $[V_{\text{vac}}^2] = G M_{\odot} \text{kpc}^{-1} = (\text{km/s})^2$. ✓

C.6 Complete chain — summary

Full derivation from Axiom 1:

Axiom 1: isotropy + max ρ_{holo} + min complexity

↓ (C.1)

$z_{\text{fund}} = 4$ [unique Platonic solid]

↓ (C.2.1)

$D_A = \ln 4 = 1.38629 \dots$

↓ (C.2.2)

$D_V = \ln 8 = 2.07944 \dots$

↓ (C.2.3) — algebraic identity

$D_V - D_A = \ln 2 = 0.69315 \dots$ [exact, $\forall z$]

↓ (C.3)

$\sigma_{\text{UV}} = 2 \ln(3/2) / \ln 12 = 0.32634 \dots$ [2 integers]

↓ (C.4.1)

$N_{\text{bits}} = \log_2 8 = 3$ [exact, $= \dim(\mathbb{R}^3)$]

↓ (C.4.3) — equipartition ansatz

$\sigma_{\text{eff}} = \sigma_{\text{UV}}/3 = 0.10878 \dots$ [derived prediction]

↓ (C.4.4) — 171 real SPARC galaxies

$\sigma_{\text{eff}}^{\text{SPARC}} = 0.108 \pm 0.005$ [confirmed, 0.72%]

D Phase 1 Results: Derivation of r_J — Exhaustive Exploration

This appendix documents the systematic exploration of six derivation paths for the Jeans radius r_J in physical units from TEG constants. All paths were tested numerically on 171 real SPARC galaxies. Documenting failed paths is part of honest science.

D.1 The problem

The vacuum entropy term currently uses a cubic profile $M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}} \cdot (r/r_{\text{max}})^3$, which gives $V_{\text{vac}} \propto r$ (rising). A flat rotation curve requires $M_{\text{vac}} \propto r$, which demands a transition radius $r_{\text{trans}} = 2r_J$ where the baryon-dominated and vacuum-dominated regimes switch. Six paths were explored to derive r_J from TEG constants without free parameters.

D.2 Six explored paths

Path 1 — Tetrahedral network cell density. If the network has cell size λ , then $n_{\text{vac}} = 24\sqrt{2}/\lambda^3$. For $\lambda = l_{\text{Planck}}$, this gives microscopic densities and $r_J \sim 10^{-20}$ kpc. Not physical at galactic scales.

Path 2 — Entropy balance $D_V \cdot \text{Vol} = D_A \cdot \text{Area}$. This gives $r_{\text{trans}} = 3D_A/D_V = 3\ln 4/\ln 8 = 2$ exactly — but dimensionless. Converting to kpc requires an external length scale not present in TEG.

Path 3 — Roughness saturation $F(N)$. The saturation function $F(N) = 1 - e^{-N/N_0}$ gives a mass scale ($M_b = M_J \cdot 10^{1.04} \approx 10^{7.04} M_\odot$) but not a radius scale.

Path 4 — Entropy ratio D_A/D_V . $r_{\text{trans}} = \ln 4/\ln 8 = 2/3$ exactly — again dimensionless. Note: $\ln 4/\ln 2 = \log_2 4 = 2$ exactly (algebraic identity). The ratio of TEG entropy constants is always rational, never a length.

Path 5 — Self-consistency $V_{\text{bar}}(r_J) = V_{\text{vac}}(r_J)$. Solving gives $r_J = r_{\text{max}} \cdot (\ln 4/\ln 2)^{1/3} = 1.2599 r_{\text{max}}$. This places r_J *outside* the observed data range for all 171 galaxies — the transition never occurs within the data, making the linear term inoperative.

Path 6 — $M_b(r_J) = M_J$ on the observed profile. Setting r_J as the radius where the enclosed baryon mass equals the Jeans mass reference gives a median $r_J = 0.58$ kpc (median $r_J/r_{\text{max}} = 0.047$). This is far too small: the flat V_{vac} amplitude becomes enormous, giving RMSE = 0.52 dex vs. 0.145 dex for the cubic.

D.3 Numerical results

D.4 Conclusion and implication

r_J in physical units requires the intrinsic cell density of the tetrahedral vacuum network. This density can only be computed from a full quantum theory of the vacuum (LQG or CDT), where the Planck-scale network parameters are fixed by the dynamics. Until then, $M_{\text{vac}} \propto r^3$ is the correct profile for TEG, using only observable quantities (r_{max} , $M_{b,\text{tot}}$) and the derived coefficient $\ln 2$.

The linear $M_{\text{vac}} \propto r$ profile remains the primary target for TEG v1.2, conditional on deriving the network cell density from QG.

Table 6: Performance of all explored M_{vac} profiles on 171 SPARC galaxies. The cubic profile (current TEG v1.1) remains optimal.

Profile	RMSE	Std	Improved	Status
Baseline (no M_{vac})	0.1617	0.081	—	Reference
$M_{\text{vac}} \propto r^3$ (cubic, current)	0.1453	0.078	127/171	Best
Flat, $r_{\text{trans}} = r_{\text{max}}$	0.1732	0.086	78/171	Worse
Flat, $r_{\text{trans}} = 2r_{\text{max}}$	0.1632	0.077	92/171	Worse
Flat, $r_{\text{trans}} = r_J$ (Path 6)	0.5234	0.193	17/171	Much worse

E Phase 2: Equipartition from Quantum Gravity

This appendix presents the strongest available argument for $\sigma_{\text{eff}} = \sigma_{\text{UV}}/3$ from first principles in loop quantum gravity (LQG) and causal dynamical triangulations (CDT). It elevates the equipartition from a bare ansatz to a physically motivated result, while stating honestly what remains unproved.

E.1 The question

Why is the denominator in $\sigma_{\text{eff}} = \sigma_{\text{UV}}/N_{\text{bits}}$ equal to 3 and not 2 or π ? The surface answer — “3 spatial dimensions” — is insufficient. We require a derivation from the structure of the quantum vacuum itself.

E.2 LQG: 4-valent nodes and minimal degeneracy

In loop quantum gravity (Rovelli & Smolin 1995 [9]), spin network nodes can have any valence n . The semiclassical sector is dominated by nodes that minimise complexity while admitting non-trivial internal structure. For spin- $\frac{1}{2}$ links, the Hilbert space of intertwiners has dimension:

Valence n	$\dim(\mathcal{H}_{\text{int}})$	Note
3	1	Only one singlet — no internal structure
4	2	Minimum with non-trivial structure
5	3	
6	5	(Fibonacci sequence)

The 4-valent node is the unique minimum with $\dim > 1$, corresponding exactly to $z_{\text{fund}} = 4$ selected by Axiom 1. Its equilibrium state under isotropy is the maximally mixed state $\rho = \mathbf{I}_2/2$.

E.3 Von Neumann entropy equals the holographic bit

The von Neumann entropy of the 4-valent node:

$$S_{\text{vN}}(\rho) = -\text{Tr}(\rho \ln \rho) = -\text{Tr}\left(\frac{\mathbf{I}_2}{2} \ln \frac{\mathbf{I}_2}{2}\right) = \ln 2. \quad (24)$$

Numerical verification: $S_{\text{vN}} = 0.693147180559945 = \ln 2$ to 15 significant figures.

This is not a coincidence. It is the same $\ln 2 = D_V - D_A$ that appears as the universal holographic bit (Appendix C, Step C.2.3). The 4-valent LQG node, the TEG bulk–surface entropy difference, and the standard Shannon bit are the same mathematical object.

E.4 Isotropy factorises the state

By Axiom 1, the vacuum is isotropic in $\mathbb{R}^3$: no spatial direction is preferred. Therefore the state of the 3-node network (encoding all three spatial directions) factorises:

$$\rho_{\text{total}} = \rho_x \otimes \rho_y \otimes \rho_z, \quad \rho_x = \rho_y = \rho_z = \frac{\mathbf{I}_2}{2}. \quad (25)$$

The total entropy:

$$S_{\text{total}} = S_x + S_y + S_z = 3 \ln 2 = D_V. \quad (26)$$

Note: $3 \ln 2 = \ln 8 = D_V$ — the bulk entropy of TEG emerges from the total von Neumann entropy of the three-direction LQG state.

E.5 Equipartition from the partial trace

The roughness σ_{UV} is a property of the total vacuum–matter interface. Under the isotropic factorisation of Eq. (25), the contribution from a single spatial direction is obtained by partial trace over the other two:

$$\rho_x = \text{Tr}_{yz}(\rho_{\text{total}}) = \frac{\mathbf{I}_2}{2}, \quad S_x = \ln 2 = \frac{S_{\text{total}}}{3} = \frac{D_V}{3}. \quad (27)$$

The fraction of the total entropy carried by one direction:

$$\frac{S_x}{S_{\text{total}}} = \frac{\ln 2}{3 \ln 2} = \frac{1}{3} = \frac{1}{N_{\text{bits}}}. \quad (28)$$

Applying this fraction to the UV roughness:

$$\boxed{\sigma_{\text{eff}} = \sigma_{\text{UV}} \cdot \frac{1}{N_{\text{bits}}} = \frac{\sigma_{\text{UV}}}{3} = 0.1088.} \quad (29)$$

The factor $1/3$ is not imposed by hand. It emerges from the partial trace of the maximally mixed state of the 4-valent LQG node over two of three isotropic spatial degrees of freedom.

E.6 CDT consistency

Ambjorn, Jurkiewicz & Loll (2005) [10] showed that in the extended de Sitter phase of CDT, the spatial volume grows as $V(t) \propto \cosh^3(t/\ell)$ — uniform distribution across all three spatial directions. This is precisely the equipartition of Eq. (25) at the macroscopic level. Furthermore, the spectral dimension in CDT falls to $D_{\text{spec}} \in [2.0, 2.5]$ at small scales, bracketing $D_{\text{eff}} = \ln 8 \approx 2.079$.

E.7 Honest assessment

What this argument proves: Under the assumptions that (i) the semiclassical vacuum is dominated by 4-valent LQG nodes with spin- $\frac{1}{2}$ links, and (ii) the equilibrium state is maximally mixed under isotropy, the equipartition $\sigma_{\text{eff}} = \sigma_{\text{UV}}/3$ follows from the partial trace.

What remains unproved: The assumption $j = \frac{1}{2}$ for all links is a simplification. The actual semiclassical state may be a superposition over spins j , modifying the intertwiner dimension and hence the von Neumann entropy. A complete proof requires computing the partition function of the tetrahedral spin network at the Jeans mass scale. This is a well-posed but unsolved problem in LQG.

Status upgrade: From *bare ansatz* (TEG v1.0) to *physically motivated result supported by LQG and CDT* (TEG v1.2). A rigorous proof remains open.

E.8 Summary of consistencies

Table 7: Numerical consistencies between TEG and quantum gravity.

Quantity	TEG value	QG prediction	Source	Match
D_{eff}	$\ln 8 = 2.079$	$[2.0, 2.2]$	LQG (Modesto 2009)	Yes
D_{eff}	$\ln 8 = 2.079$	$[2.0, 2.5]$	CDT (Ambjorn 2005)	Yes
$S_{\text{vN}}(4\text{-valent node})$	$\ln 2$	$\ln 2$	LQG intertwiners	Exact
N_{bits}	3 (exact)	3 spatial dim. (CDT de Sitter)	CDT (2005)	Yes
σ_{eff}	0.1088 (derived)	0.108 ± 0.005	171 SPARC galaxies	0.72%